Poisson Distribution

ANSWER KEY

Section 1: Poisson Probability Formula & Conditions

1 Which of the following is NOT an essential assumption for modeling a random variable with a Poisson distribution $\text{Po}(\lambda)$?

a. **Events occur at a variable rate over time** ✓

b. Events occur singly and independently

c. The average rate of occurrence λ is constant

d. Two events cannot occur at the exact same instant

In a Poisson process, events must occur at a constant average rate, independently, at random, and singly.

2 The random variable $X \sim \text{Po}(3.2)$. Find $P(X = 2)$ correct to 4 decimal places.

a. 0.2087 ✓

b. 0.2226

c. 0.1782

d. 0.1450

$P(X = 2) = \frac{e^{-3.2}(3.2)^2}{2!} = \frac{0.0407622 \times 10.24}{2} = 0.2087.$

3 If $X \sim \text{Po}(\lambda)$ and $P(X = 0) = 0.05$, find the value of λ.

a. $\lambda = -\ln(0.05) \approx 2.996$ ✓

b. $\lambda = 0.05$

c. $\lambda = e^{0.05} \approx 1.051$

d. $\lambda = 20$

$P(X = 0) = e^{-\lambda} = 0.05 \Rightarrow -\lambda = \ln(0.05) \Rightarrow \lambda = -\ln(0.05) \approx 2.996.$

Section 2: Mean, Variance & Linear Combinations

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4 What is the theoretical relationship between the mean and variance of a Poisson distribution $X \sim \text{Po}(\lambda)$?

a. **Mean** $= \text{Variance} = \lambda$ ✓

b. Mean $= \lambda, \text{Variance} = \lambda^2$

c. Mean $= \lambda, \text{Variance} = \sqrt{\lambda}$

d. Mean $= \lambda, \text{Variance} = \lambda(1-p)$

A defining property of the Poisson distribution is that its mean and variance are equal: $E(X) = \text{Var}(X) = \lambda$.

5 Independent random variables $X \sim \text{Po}(2)$ and $Y \sim \text{Po}(3)$ represent customer arrivals. What distribution does $X + Y$ follow?

a. $\text{Po}(5)$ ✓

b. $\text{Po}(6)$

c. $\text{Po}(2.5)$

d. $N(5, 5)$

The sum of independent Poisson variables is also Poisson: $X + Y \sim \text{Po}(\lambda_X + \lambda_Y) = \text{Po}(2 + 3) = \text{Po}(5)$.

6 Faults on a fiber optic cable occur at an average rate of 0.4 per kilometer. What is the probability of finding exactly 1 fault in a 5-kilometer section?

a. 0.2707 ✓

b. 0.4000

c. 0.1353

d. 0.3250

Rate for 5 km is $\lambda = 0.4 \times 5 = 2.0$. Then $P(X = 1) = \frac{e^{-2}(2)^1}{1!} = 2e^{-2} = 2(0.135335) \approx 0.2707$.

Section 3: Poisson Approximation to the Binomial

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7 Under which conditions can a binomial distribution $B(n, p)$ be well approximated by a Poisson distribution?

a. $n > 50$ **and** $np < 5$ ✓

b. $n > 30$ and $p = 0.5$

c. $np > 5$ and $n(1 - p) > 5$

d. $n < 20$ and $p > 0.1$

The Poisson approximation to the binomial is appropriate when the number of trials n is large ($n > 50$) and the probability of success p is small ($np < 5$), with parameter $\lambda = np$.

8 A book of 400 pages contains 200 typographical errors distributed at random. Use a Poisson approximation to find the probability that a randomly chosen page contains no errors.

a. $e^{-0.5} \approx 0.6065$ ✓

b. 0.5000

c. 0.3679

d. 0.7500

Average errors per page $\lambda = 200/400 = 0.5$. $P(X = 0) = e^{-0.5} \approx 0.6065$.

9 A factory produces screws, of which 1% are defective. In a random sample of 250 screws, what is the Poisson parameter λ to model defective screws?

a. $\lambda = 2.5$ ✓

b. $\lambda = 0.01$

c. $\lambda = 25$

d. $\lambda = 5$

$\lambda = np = 250 \times 0.01 = 2.5$.

Section 4: Normal Approximation to the Poisson Distribution

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10 Under what condition can a Poisson distribution $X \sim \text{Po}(\lambda)$ be approximated by a normal distribution?

a. $\lambda > 15$ ✓

b. $\lambda < 5$

c. λ is an integer

d. $\lambda > 100$ only

When $\lambda > 15$, the distribution is sufficiently symmetric to be approximated by $N(\lambda, \lambda)$.

11 If $X \sim \text{Po}(36)$, which normal distribution provides the best approximation?

a. $N(36, 36)$ **with mean 36 and variance 36** ✓

b. $N(36, 6)$

c. $N(0, 1)$

d. $N(36, 18)$

Since mean and variance of $\text{Po}(\lambda)$ are both λ, the approximating normal distribution is $N(\lambda, \lambda) = N(36, 36)$.

12 For $X \sim \text{Po}(36)$, use the normal approximation with continuity correction to state the calculation for $P(X \leq 40)$.

a. $P(Y < 40.5)$ **where** $Y \sim N(36, 36)$ ✓

b. $P(Y < 40)$ where $Y \sim N(36, 36)$

c. $P(Y < 39.5)$ where $Y \sim N(36, 36)$

d. $P(Y < 41)$ where $Y \sim N(36, 6)$

For discrete $X \leq 40$, the continuity correction extends to the upper boundary 40.5: $P(Y < 40.5)$.

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